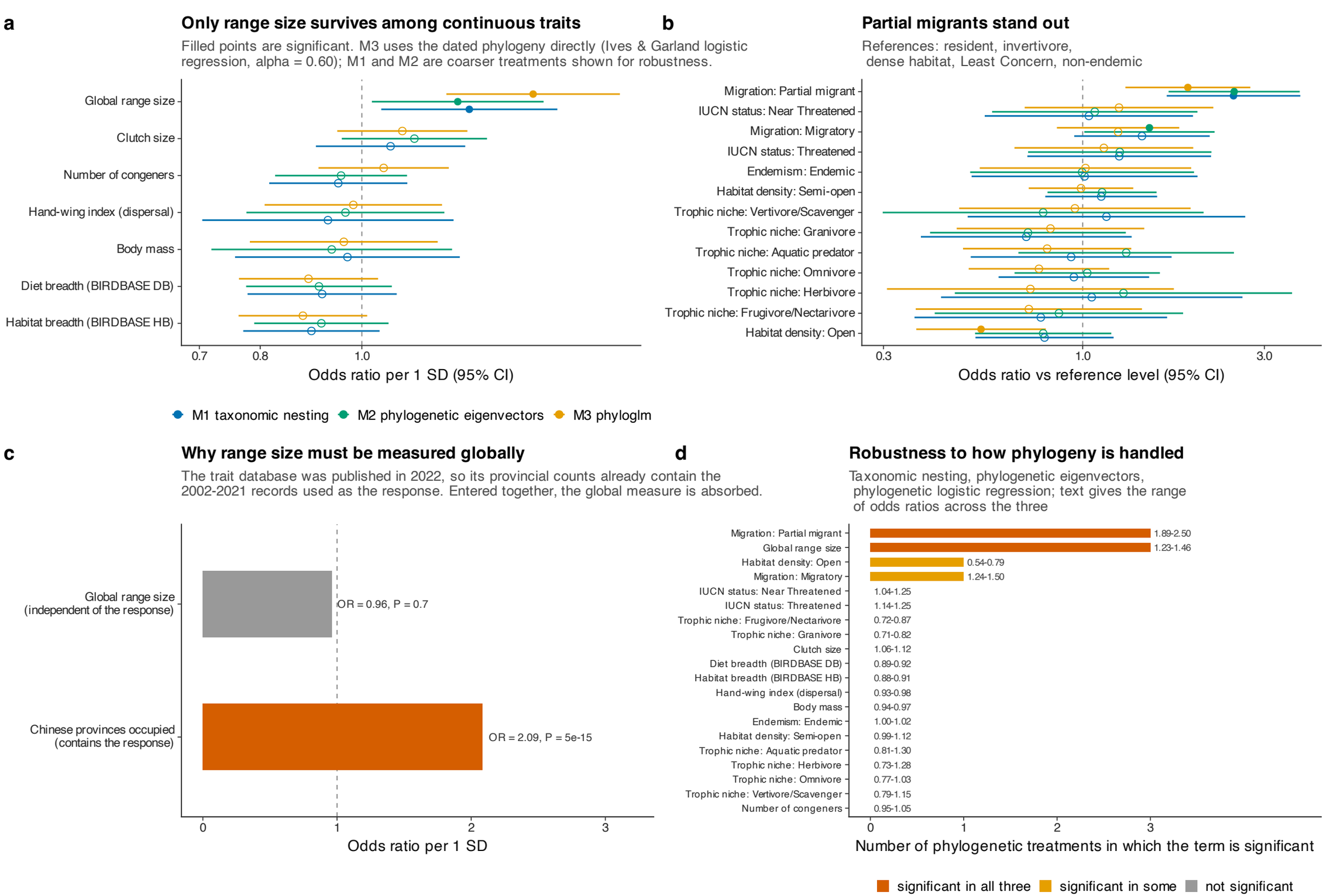
a
Only range size survives among continuous traits
Filled points are significant. M3 uses the dated phylogeny directly (Ives & Garland logistic regression, alpha = 0.60); M1 and M2 are coarser treatments shown for robustness.
Global range size
Clutch size
Number of congeners
Hand-wing index (dispersal)
Body mass
Diet breadth (BIRDBASE DB)
Habitat breadth (BIRDBASE HB)
0.7
0.8
1.0
Odds ratio per 1 SD (95% CI)
M1 taxonomic nesting
M2 phylogenetic eigenvectors
M3 phyloglm
b
Partial migrants stand out
References: resident, invertivore, dense habitat, Least Concern, non-endemic
Migration: Partial migrant
IUCN status: Near Threatened
Migration: Migratory
IUCN status: Threatened
Endemism: Endemic
Habitat density: Semi-open
Trophic niche: Vertivore/Scavenger
Trophic niche: Granivore
Trophic niche: Aquatic predator
Trophic niche: Omnivore
Trophic niche: Herbivore
Trophic niche: Frugivore/Nectarivore
Habitat density: Open
0.3
1.0
3.0
Odds ratio vs reference level (95% CI)
c
Why range size must be measured globally
The trait database was published in 2022, so its provincial counts already contain the 2002-2021 records used as the response. Entered together, the global measure is absorbed.
Global range size (independent of the response)
OR = 0.96, P = 0.7
Chinese provinces occupied (contains the response)
OR = 2.09, P = 5e-15
0
1
2
3
Odds ratio per 1 SD
d
Robustness to how phylogeny is handled
Taxonomic nesting, phylogenetic eigenvectors, phylogenetic logistic regression; text gives the range of odds ratios across the three
Migration: Partial migrant 1.89-2.50
Global range size 1.23-1.46
Habitat density: Open 0.54-0.79
Migration: Migratory 1.24-1.50
IUCN status: Near Threatened 1.04-1.25
IUCN status: Threatened 1.14-1.25
Trophic niche: Frugivore/Nectarivore 0.72-0.87
Trophic niche: Granivore 0.71-0.82
Clutch size 1.06-1.12
Diet breadth (BIRDBASE DB) 0.89-0.92
Habitat breadth (BIRDBASE HB) 0.88-0.91
Hand-wing index (dispersal) 0.93-0.98
Body mass 0.94-0.97
Endemism: Endemic 1.00-1.02
Habitat density: Semi-open 0.99-1.12
Trophic niche: Aquatic predator 0.81-1.30
Trophic niche: Herbivore 0.73-1.28
Trophic niche: Omnivore 0.77-1.03
Trophic niche: Vertivore/Scavenger 0.79-1.15
Number of congeners 0.95-1.05
0
1
2
3
Number of phylogenetic treatments in which the term is significant
significant in all three
significant in some
not significant